

Summary

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The First Fundamental Theorem of Calculus:

Given a continuous function f on the interval $[a, b]$, we define an accumulation function, A by

$$A(x) = \int_a^x f(t) dt$$

The First Fundamental Theorem of Calculus (FFTOC) states that

$$A'(x) = f(x)$$

or, equivalently, that

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Note 1: $A(a) = 0$

Note 2: The function A is an antiderivative of f .

Therefore, any antiderivative, G of f can be written as $G(x) = A(x) + C$, or, equivalently,

$$G(x) = \int_a^x f(t) dt + C.$$

Since $G(a) = A(a) + C = 0 + C = C$, it follows that $G(x) = A(x) + G(a)$, or, equivalently, that

$$G(x) = G(a) + \int_a^x f(t) dt$$

Note 3: Let f' , the derivative of f , be continuous on $[a, b]$. Since f is its antiderivative, we have

$$f(x) = f(a) + \int_a^x f'(t) dt$$

The Second Fundamental Theorem of Calculus:

FFTOC implies that, for $x = b$, $A(b) = \int_a^b f(t) dt$. Since $A(a) = 0$, we can write

$$\int_a^b f(t) dt = A(b) - A(a)$$

Note 4: This is the Second Fundamental Theorem of Calculus (SFTOC).

Note 5: Equivalently, using the result in Note 3, for $x = b$, $f(b) = f(a) +$

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$\int_a^b f'(t) dt$. So $\int_a^b f'(t) dt = f(b) - f(a)$, or equivalently,

$$\int_a^b \frac{d}{dx} f(x) dx = f(b) - f(a).$$